ASSIGNMENT 1 – CO1 | TASK 2

Tokens, Keywords, Identifiers and Constants

Scenario: Namma Metro Smart-Card Fare System

Objective / explanation

This task defines fares and rules in one place using constexpr constants and a scoped enum class. This makes revisions easier and safer than repeating numeric literals.

C++ source code

```
#include &lt;iostream&gt;
using namespace std;

enum class Zone { Inner, Middle, Outer };
constexpr long long BASE_PAISE[] = {1000, 2000, 3000};
constexpr int PEAK_PERCENT = 20;
constexpr long long MIN_BALANCE_PAISE = 5000;

int main() {
    cout &lt;&lt; "Inner fare paise: " &lt;&lt; BASE_PAISE[0] &lt;&lt; '\n';
    cout &lt;&lt; "Peak surcharge percent: " &lt;&lt; PEAK_PERCENT &lt;&lt; '\n';
    cout &lt;&lt; "Minimum balance paise: " &lt;&lt; MIN_BALANCE_PAISE &lt;&lt; '\n';
}
```

Compilation

```
g++ -std=c++17 -Wall -Wextra -pedantic main.cpp -o program
```

Actual captured program output

```
Inner fare paise: 1000
Peak surcharge percent: 20
Minimum balance paise: 5000
```

Output screenshot

Analysis / complexity

Keywords include enum and constexpr; identifiers include BASE_PAISE and PEAK_PERCENT; operators include << and =; punctuators include braces and semicolons. constexpr values are type-checked.

Evidence note: Output shown is captured by executing this source in the generation environment. Recompile and run on your own system, and use your own screenshots if your instructor requires evidence from your computer.